

### Lesson 3 Logical Equivalence

In Lesson 2, we focused on the precise meaning of connectives used in math statements and touched upon the concept of **logical equivalence**:  $p \leftrightarrow q$  was defined as being *equivalent* to “ $p \rightarrow q$  and  $q \rightarrow p$ ”.

- Statements  $r$  and  $s$  are (logically) **equivalent**, denoted  $r = s$ , if  $r$  and  $s$  have the same meaning. Logically equivalent statements have the *same truth table*.

In this lesson, we discuss a set of basic equivalences. Let  $p$ ,  $q$ , and  $r$  be statements.

**(Commutative)**  $p \wedge q = q \wedge p$ ,  $p \vee q = q \vee p$

**(Associative)**  $(p \wedge q) \wedge r = p \wedge (q \wedge r)$ ,  $(p \vee q) \vee r = p \vee (q \vee r)$

**(Distributive)**  $p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$ ,  $p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$

These logic laws can be proved using truth table (exercises). They have familiar arithmetic counterparts:  $a + b = b + a$ ,  $ab = ba$ ,  $a + (b + c) = (a + b) + c$ ,  $a(bc) = (ab)c$ , and  $a(b + c) = ab + ac$ .

**(Idempotent)**  $p \wedge p = p$ ,  $p \vee p = p$

**(Domination)**  $p \vee T = T$ ,  $p \wedge F = F$

**(Identity)**  $p \wedge T = p$ ,  $p \vee F = p$

**(Negation)**  $p \wedge \sim p = F$ ,  $p \vee \sim p = T$

**(Absorption)**  $p \vee (q \wedge p) = p$ ,  $p \wedge (q \vee p) = p$

The first four laws follow readily from the truth table of  $\wedge$  and  $\vee$ . For example,  $T \wedge T = T$  and  $F \wedge F = F$ , hence the idempotent law:  $p \wedge p = p$ . An idempotent is an element that is unchanged when operated by itself. For example, 0 and 1 are idempotents under multiplication. Next, one of  $p$  and  $\sim p$  is true, so  $p \vee \sim p$  is true (negation law). A statement which is always true *due to its logical form*, such as  $p \vee \sim p$ , is a **tautology**, and one which is always false is a **contradiction**. A tautology is true, “it works or doesn’t”, but gives no information. Not all true statements are tautology; Pythagoras’ theorem is true, but it is not a tautology, it gives plenty of information! The absorption laws are proved in the exercises.

**(Double negatives)**  $\sim(\sim p) = p$

Let  $p$  be “it rains”, then  $\sim p$  is “it does not rain” and  $\sim(\sim p)$  is “it rains”.

| $p$ | $\sim p$ | $\sim(\sim p)$ |
|-----|----------|----------------|
| T   | F        | T              |
| F   | T        | F              |

**(De Morgan’s Laws)**  $\sim(p \wedge q) = \sim p \vee \sim q$ ,  $\sim(p \vee q) = \sim p \wedge \sim q$

| $p$ | $q$ | $p \wedge q$ | $p \vee q$ | $\sim(p \wedge q)$ | $\sim(p \vee q)$ | $\sim p$ | $\sim q$ | $\sim p \vee \sim q$ | $\sim p \wedge \sim q$ |
|-----|-----|--------------|------------|--------------------|------------------|----------|----------|----------------------|------------------------|
| T   | T   | T            | T          | F                  | F                | F        | F        | F                    | F                      |
| T   | F   | F            | T          | T                  | F                | F        | T        | T                    | F                      |
| F   | T   | F            | T          | T                  | F                | T        | F        | T                    | F                      |
| F   | F   | F            | F          | T                  | T                | T        | T        | T                    | T                      |

The negation of “ $x$  is prime *and* even” is “ $x$  is not prime *or*  $x$  is not even”. The negation of  $p \wedge q$  is NOT  $\sim p \wedge \sim q$ :  $\sim(p \wedge q)$  and  $\sim p \wedge \sim q$  do not have the same truth table, or we may just give a case where they differ: let  $p = T$  and  $q = F$ , then  $p \wedge q = F$  and  $\sim(p \wedge q) = T$ , but  $\sim p = F$ ,  $\sim q = T$ , and  $\sim p \wedge \sim q = F$ .

**(Contrapositive)**  $p \rightarrow q = \sim q \rightarrow \sim p$ .

“If  $x$  is even, then the square of  $x$  is even” is equivalent to its contrapositive, “if the square of  $x$  is not even, then  $x$  is not even.” The conditional  $\sim q \rightarrow \sim p$  is the *contrapositive* of  $p \rightarrow q$ . We can prove the contrapositive equivalence with a truth table (exercise), but it is also intuitive: since  $p$  implies  $q$ , if “not  $q$ ”, then “not  $p$ ”! So,  $p$  is **sufficient** for  $q$ , and  $q$  is a **necessary** for  $p$ . Let  $x$  and  $y$  be integers. Consider the true statement “if  $x \geq 1$ , then  $x^2 \geq 1$ ”:  $x \geq 1$  is sufficient but not necessary for  $x^2 \geq 1$ :  $2 < 1$  but  $(-2)^2 \geq 1$ . However,  $x^2 \geq 1$  is necessary for  $x > 1$ : if  $x^2 < 1$ , then we cannot have  $x \geq 1$ . The inverse of  $p \rightarrow q$ ,  $\sim p \rightarrow \sim q$ , is equivalent to the converse  $q \rightarrow p$  as  $\sim p \rightarrow \sim q$  is the contrapositive of  $q \rightarrow p$ .

**(Conditional as “or”)**  $p \rightarrow q = \sim p \vee q$

| $p$ | $q$ | $p \rightarrow q$ | $\sim p$ | $\sim q$ | $\sim p \vee q$ |
|-----|-----|-------------------|----------|----------|-----------------|
| T   | T   | T                 | F        | F        | T               |
| T   | F   | F                 | F        | T        | F               |
| F   | T   | T                 | T        | F        | T               |
| F   | F   | T                 | T        | T        | T               |

For example, “if an integer is not even, then it is odd” is equivalent to “an integer is even or odd”. That a conditional can be written as an “or” statement is very significant and useful, it enables us to negate a conditional!

**(Negation of Conditional)**  $\sim(p \rightarrow q) = p \wedge \sim q$

We could use a truth table to prove, but we will use *proven equivalence laws*:

$$\begin{aligned}
 \sim(p \rightarrow q) &= \sim((\sim p) \vee q) && \text{Conditional as “or”} \\
 &= \sim(\sim p) \wedge \sim q && \text{De Morgan} \\
 &= p \wedge \sim q && \text{Double Negatives}
 \end{aligned}$$

The negation of the conditional “if an integer is not even, then it is odd” is “an integer is not even, and it is not odd.” The negation of  $p \rightarrow q$  is NOT  $p \rightarrow \sim q$  or any conditional, this is one of the most common mistakes students make!

**(“and” implies “or”)** The following are tautologies:

$$(p \wedge q) \rightarrow p, \quad (p \wedge q) \rightarrow q, \quad p \rightarrow (p \vee q), \quad q \rightarrow (p \vee q), \quad (p \wedge q) \rightarrow (p \vee q)$$

We can prove the above tautologies using truth table or earlier laws, but these tautologies are also intuitive: if we have both  $p$  and  $q$ , then we have  $p$  (or  $q$ ) in particular. If  $p$  is true, then  $p$  **or**  $q$  is true, whatever  $q$  is, but be careful,  $p$  does NOT imply  $p$  and  $q$ !